

Wireshark & Packet Analysis

Why Packet Analysis

The network is the attacker's narrative. Traffic analysis shows you:

- What services are talking
- Whether encryption is used
- Whether data exfiltration is happening
- Malware C2 beacons, DNS tunneling, brute force, scans

Capture Concepts

Wireshark captures frames on an interface. Capture in **promiscuous mode** to see all traffic on the wire – but on switched networks you typically only see your own (or mirrored) traffic.

Capture files: **PCAP** (`.pcap`, `.pcapng`). Investigators analyze PCAPs that were captured by sensors, not by themselves – that's true network forensic work.

Essential Filters

Filter	Shows
<code>ip.addr == 192.168.1.50</code>	Traffic to/from an IP
<code>tcp.port == 80</code>	HTTP-only traffic
<code>http.request</code>	HTTP GET/POST requests
<code>dns</code>	DNS queries
<code>tcp.flags.syn == 1</code>	SYN packets (scans!)
<code>ftp-data</code>	FTP file transfers

Filter	Shows
<code>icmp</code>	Pings
<code>frame contains "password"</code>	Raw payload search

Following TCP Streams

Right-click a packet → **Follow** → **TCP Stream**. Wireshark reassembles the conversation – you can read the entire HTTP request/response including credentials in cleartext if not TLS-encrypted.

This is how MFAs get bypassed in labs and how SOC analysts confirm data exfiltration.

Analyzing Captures for Attacks

Scan detection: Many SYN packets without SYN-ACK replies → a scan.

Filter: `tcp.flags.syn == 1 and tcp.flags.ack == 0`

Brute force: Repeated 401 Unauthorized responses or repeated failed logins → dictionary attack.

C2 beaconing: Regular-interval packets to the same IP (e.g., every 60s) of the same size.

Data exfil: Large outbound transfers on unusual ports, base64-looking payloads.

Practical Triage Workflow

Statistics → **Endpoints** – who's talking?

Statistics → **Conversations** – main volume pairs

Protocol Hierarchy – what protocols, in what volume?

Filter on anomalies → Follow streams → Document

← HTTP, HTTPS & TLS Deep Dive

Firewalls, IDS & IPS →